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Successful Well Delivery with Early Engagement and Real-Time Geomechanics Support within a Heterogeneous Carbonate Reservoir

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Abstract

This paper presents the successful execution of a technically demanding lateral well, drilled through a geologically complex and high-risk reservoir. The well trajectory was deliberately oriented in the direction of the minimum horizontal stress (S_{hmin}) within a strike-slip tectonic regime—a configuration known to pose significant wellbore stability challenges. The complexity of the operation was further intensified by pronounced reservoir heterogeneity, which introduced substantial variations in both rock mechanical properties and in-situ stress profiles along the lateral section.

Compounding these geological challenges, the well was classified as a high-temperature environment, which imposed strict constraints on the hydraulic regime. These thermal limitations restricted the operational flexibility of mud weight and flow rates, making it critical to strike a precise balance between wellbore stability and hydraulic safety margins.

To address these multifaceted challenges, a robust pre-drill geomechanical model was developed, incorporating detailed offset well analysis and accounting for reservoir depletion effects and stress redistribution. This model enabled the definition of a narrow but reliable mud weight window, tailored to the specific geomechanical conditions of the reservoir. The mud weight strategy was further reinforced by the deployment of full Real-Time Geomechanics support throughout the drilling phase, ensuring continuous monitoring and rapid response to any emerging instability risks.

As part of a proactive, early engagement strategy, the geomechanical study and its findings were presented to both the client and Baker Hughes teams during the planning phase. This collaborative, multidisciplinary approach ensured alignment on risk mitigation strategies and operational readiness. During drilling, a dedicated real-time advisory group continuously updated the geomechanical model using live downhole data and provided timely, data-driven recommendations to the drilling team.

Drilling horizontal wells in strike-slip regimes—especially in the S_{hmin} direction—is inherently risky due to the increased likelihood of borehole breakouts and instability. These risks are further exacerbated by lateral variations in reservoir properties, which introduce uncertainty into pre-drill models and complicate real-time decision-making. In this context, real-time geomechanics monitoring proved essential for maintaining borehole integrity and minimizing Non-Productive Time (NPT).

The lateral section was drilled entirely within the recommended mud weight envelope derived from the pre-drill model. The operation was completed as planned, with no significant wellbore instability events and no recordable NPT. The result was a smooth, in-gauge wellbore that facilitated the acquisition of high-quality wireline logs, enabling accurate reservoir evaluation. Following successful logging, the liner was run and cemented without incident, marking the completion of a technically successful operation.

This case study underscores the critical value of early geomechanical engagement, collaborative planning, and real-time monitoring in delivering complex wells. The seamless integration of pre-drill modeling with real-time geomechanics support was instrumental in achieving a stable wellbore and ensuring the safe and efficient execution of both drilling and completion phases.

Introduction & Background

This paper presents a compelling case study from one of the major oil and gas fields in the Middle East, where the integration of accurate pre-drill GeoMechanical and Wellbore Stability modeling with comprehensive Real-Time Geomechanics support enabled the successful drilling of a critical horizontal slim hole. The well was planned through a heterogeneous carbonate reservoir and oriented toward the minimum horizontal stress (Shmin) direction within a strike-slip tectonic regime—an orientation historically associated with significant wellbore stability challenges.

Previous drilling campaigns in this field had frequently encountered severe non-productive time (NPT) while drilling horizontal wells in similar geological settings. Common issues included tight hole conditions, hard reaming, pack-offs, and other instability-related events, particularly when drilling toward the Shmin direction. These challenges were further exacerbated by the reservoir's heterogeneity, which introduced substantial variations in rock strength and in-situ stress profiles along the lateral.

A key additional risk was the potential for differential sticking, especially in zones characterized by high porosity and permeability. In such zones, elevated hydrostatic overbalance can significantly narrow the safe mud weight window, increasing the likelihood of sticking and other drilling hazards. Compounding these risks, the lateral section was located in a high-temperature environment, which imposed strict limitations on the ability to vary the hydraulic regime during drilling.

To address this complex set of challenges, a precise pre-drill GeoMechanical model and Wellbore Stability analysis were developed. These models were used to define an optimized mud weight window tailored to the specific stress and rock property variations expected along the lateral. However, any pre-drill GeoMechanical model comes with its own set of uncertainties. Even in an ideal case, where comprehensive data sets are available, there will always be a remaining uncertainty within each of the GeoMechanical model parameters. Rock properties, pore pressure, stress magnitudes and stress azimuth come with their individual uncertainties and can affect the modelled mud weight window -in a worst-case scenario- significantly enough to cause potential drilling challenges. Thus, for further optimization of those models, Real-Time GeoMechanics services were deployed in order to update the models while drilling. During Real-Time GeoMechanics support, the GeoMechanical model and the Wellbore Stability analysis are updated using the LWD logs and drilling data of the well that is being drilled along its planned trajectory. We would like to emphasize, that for any pre-drill GeoMechanical model, naturally, only data from nearby wells can be used, and not any from the location of the planned trajectory. Hence, the rock properties, pore pressure, stress magnitudes and stress azimuth are representative only for the location of those wells from where the data was collected to build the pre-drill GeoMechanical model. This is why there is always a certain degree of uncertainty introduced to any pre-drill GeoMechanical model and its Wellbore Stability analysis, and this is why Real-Time GeoMechanics monitoring, and Real-Time GeoMechanics updates are introduced to diminish such uncertainties. In that sense, the importance of the Real-Time GeoMechanics services is a function of the complexity and variability of the Geology and with that, the complexity of the GeoMechanics: the more complex a Geology setting is (and the less data is available) the more important

the Real-Time update of the GeoMechanical model and the Wellbore Stability becomes. Part of the Real-Time GeoMechanical update is the continuous monitoring, proactive communication, and timely decision-making throughout the drilling operation. This combination of updates and communication is what makes the implementation of this process so efficient and successful.

The results in this example case that we are discussing in this publication were highly outstanding. Drilling was completed without any wellbore instability-related NPT. The GeoMechanical model was updated while drilling, though the overall Wellbore Stability model did not need any update. Thus, the mud weight recommendations could stay as originally planned. The borehole remained smooth and in-gauge throughout the entire lateral section, enabling the acquisition of high-quality wireline logs for reservoir evaluation. All planned data acquisition and completion operations—including the successful setting and cementing of a 15,000 psi-rated liner—were executed without interrupting challenges or GeoMechanics related issues.

This case highlights the critical importance of early geomechanical engagement, integrated planning, and real-time surveillance in overcoming complex drilling challenges. The collaborative approach between the operator and service provider ensured that uncertainties were minimized, risks were mitigated, and operational objectives were achieved efficiently and safely.

Predrill GeoMechanical Modeling

As the initial phase of the 1-D GeoMechanics workflow, a comprehensive review of several relevant offset wells was conducted. This step aimed to analyze historical drilling events to better understand and characterize the frequency, severity, and underlying causes of wellbore instability challenges encountered in the area. The analysis focused on identifying key indicators of instability, including but not limited to: tight hole conditions during connections and tripping operations, excessive reaming requirements, and occurrences of stuck pipe—both mechanical and differential. These events not only led to direct Non-Productive Time (NPT) but also contributed to Invisible Lost Time (ILT), such as repeated tripping operations for hole cleaning or difficulties during wireline logging that hindered essential data acquisition.

Following this diagnostic phase, the next step involved estimating the vertical stress (overburden stress). This was achieved by analyzing density logs from the offset wells. In cases where actual density logs were unavailable, pseudo-density logs were generated using the Gardner equation applied to available acoustic data. These pseudo-densities were then integrated with any existing real density data. An exponential trend line was fitted to the composite data set and extrapolated to the surface to ensure continuity. The resulting composite density logs were used to calculate the vertical stress profile across the entire depth of the well.

To determine the lithology, a combined analysis approach of Gamma Ray, Neutron Porosity, and Bulk Density logs was used. These logs were integrated to classify rock types and stratigraphy. Subsequently, elastic rock properties—such as Young's modulus and Poisson's ratio—were derived from the basic input logs (density, acoustic, neutron, etc.). These properties were calculated using industry-standard empirical correlations. These correlations were chosen to be a function of the lithology. Location-, and formation specific equations were then used to derive the static rock properties from those elastic rock properties. The static rock property equations were then further adjusted and calibrated against available lithology specific and rock specific core measurements from the region to enhance accuracy.

Traditional porosity- and effective stress-based pore pressure modeling techniques, which are typically effective in shale-dominated formations, proved inadequate in this carbonate-rich environment due to the complex and heterogeneous nature of the porosity development. Therefore, regional experience and insights from previous modeling efforts were leveraged to constrain pore pressure estimates in the shallower sections. For the reservoir section, pore pressure measurements of wells of the same field and formation were considered as well as the expected formation pressure ranges—both low-case and high-case scenarios

—as provided by the operator. These values were used to construct a final reservoir pressure profile that reflects the most likely pressure conditions.

To constrain the minimum horizontal stress (S_{hmin}), closure pressure data obtained during hydraulic fracturing operations in nearby fields was utilized, such as mini-frac and data-frac closure pressures. This data helped establish the effective stress ratio. The orientation of the maximum horizontal stress (S_{Hmax}) was inferred through a combination of detailed image log analysis from offset wells and regional stress data from the World Stress Map (WSM). However, it was acknowledged that local variations in stress azimuth near the wellbore could introduce some uncertainty due to data limitations. The magnitude of S_{Hmax} was further constrained using the observed breakout widths, the known value of S_{hmin} , and rock mechanical properties at the corresponding depths, applying the principles of frictional faulting theory. The relative magnitudes of the three principal stresses indicated that the reservoir section is situated within a strike-slip faulting stress regime.

Once the foundational 1-D GeoMechanical models for all offset wells were established, multiple calibration iterations were performed to ensure the models were representative and reliable for predictive use. These iterations involved simulating wellbore failure under the actual mud weights used during drilling of the offset wells. The predicted severity and extent of breakouts were then compared with real-world observations from image logs, caliper data and compared to observed GeoMechanics related drilling challenges while drilling those wells, such as tight holes, stuck pipe, pack-off, etc. GeoMechanical parameters were fine-tuned accordingly to achieve a close match. The resulting calibrated models were then extended along the planned trajectory of the target well to develop a predictive GeoMechanical model tailored to the new well.

This predictive model enabled the execution of multiple Wellbore Stability Analyses (WBS) to evaluate various mud weight scenarios. Among these, the most optimized and safe mud weight window was identified. This window is visually represented in [Figure 1](#), which includes lower hemisphere stereographic projection plots. These plots use color coding to indicate safe mud weight ranges—red for higher values and blue for lower values. The planned lateral section of the target well is marked with white circles for reference.

Based on this analysis, the operator elected to initiate drilling operations using the base case mud weight, which corresponds to the minimum safe mud weight of 12.3 ppg.

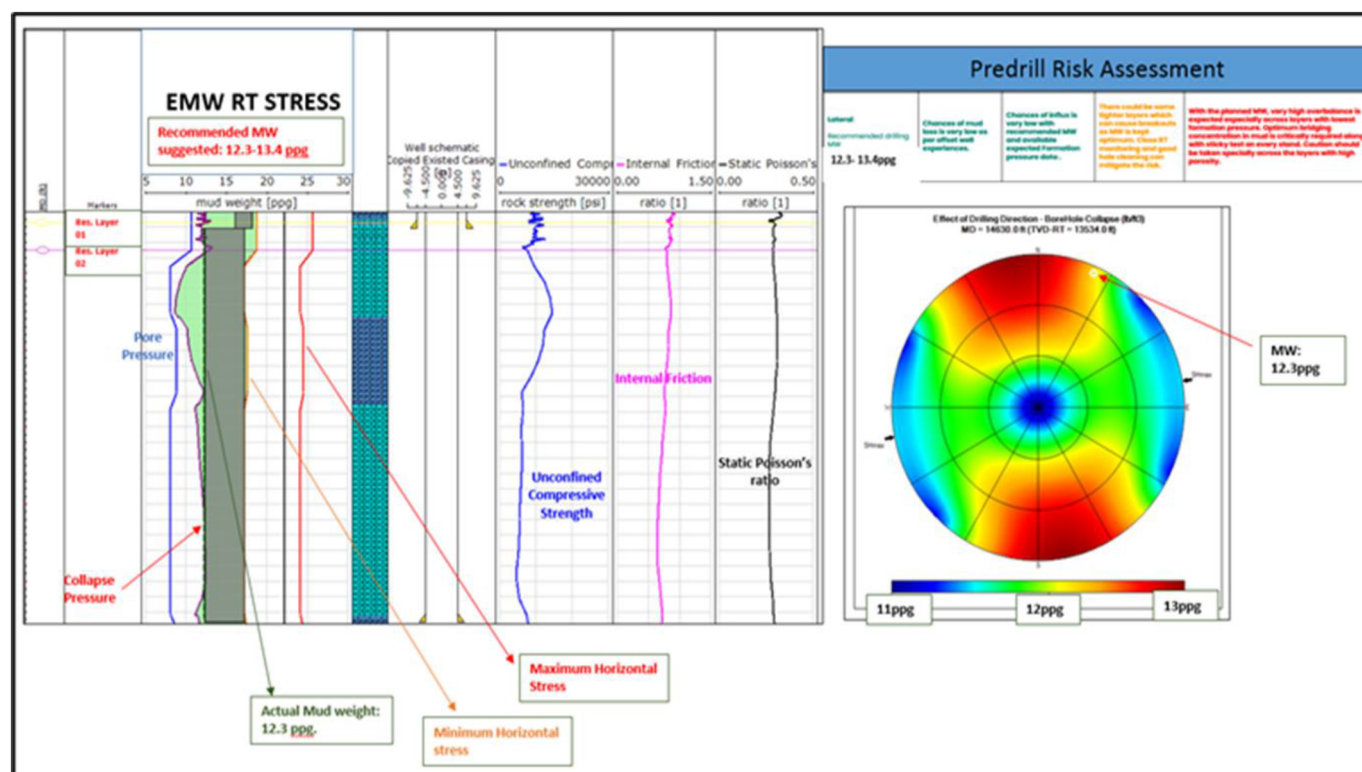


Figure 1—Summary plot showing pore pressure, stresses, rock properties and predicted mud windows vs actual mud weight for the lateral. – for the representative hole section.

Role of Real-Time GeoMechanics Support

To proactively mitigate potential wellbore stability challenges—particularly those arising from reservoir heterogeneity along the extended lateral section, which can influence both rock mechanical properties and in-situ stress conditions—Real-Time GeoMechanical services were deployed throughout the drilling operation. These services were critical in enabling continuous monitoring and timely decision-making.

A comprehensive suite of Logging While Drilling (LWD) tools was recommended and utilized, including Gamma Ray, Density, Neutron, Acoustic Caliper, Azimuthal Density Imaging, and Equivalent Circulating Density (ECD) tools. Each of these tools played a distinct and complementary role in the real-time geomechanical surveillance strategy:

- **Gamma Ray, Density, and Neutron logs** were employed to identify lithological variations, which are key indicators of changes in mechanical properties and potential zones of weakness.
- **Acoustic Caliper and Azimuthal Density Imaging tools** were instrumental in detecting the earliest signs of wellbore instability, such as borehole breakouts and enlargements, which often precede more severe failure mechanisms.
- **The ECD tool** served as a critical indicator of annular loading, particularly in the context of excessive cuttings or cavings, which can signal inefficient hole cleaning or early signs of instability.

Along with these, drilling parameters such as torque, drag, and rate of penetration were closely monitored to detect subtle changes in the formation that might indicate emerging wellbore instability. The integration of these datasets enabled a holistic view of downhole conditions, allowing the drilling team to make proactive, data-driven decisions in real time.

Drilling commenced with a mud weight of 12.3 ppg, which was the base case recommendation from the pre-drill geomechanical model. This mud weight was maintained throughout the operation due to constraints

in increasing it—primarily to avoid the risk of differential sticking and due to the limited flexibility in adjusting hydraulic parameters in a high-temperature well environment.

To ensure continuous optimization of the drilling strategy, the 1-D geomechanical model was dynamically updated during drilling. Any observed anomalies—such as breakout signatures, caliper enlargements, lithological transitions, or changes in porosity (which may indicate higher permeability and increased sticking risk)—were immediately flagged. These observations were communicated to the operator through Real-Time Geomechanical Reports, regular teleconferences, and other collaborative channels, enabling timely remedial actions.

This integrated approach—combining robust pre-drill modeling, real-time surveillance, and continuous communication—was key to maintaining wellbore integrity and avoiding major instability-related NPT throughout the drilling of the long lateral section. Figure 2 shows a snapshot of monitored parameters as an example.

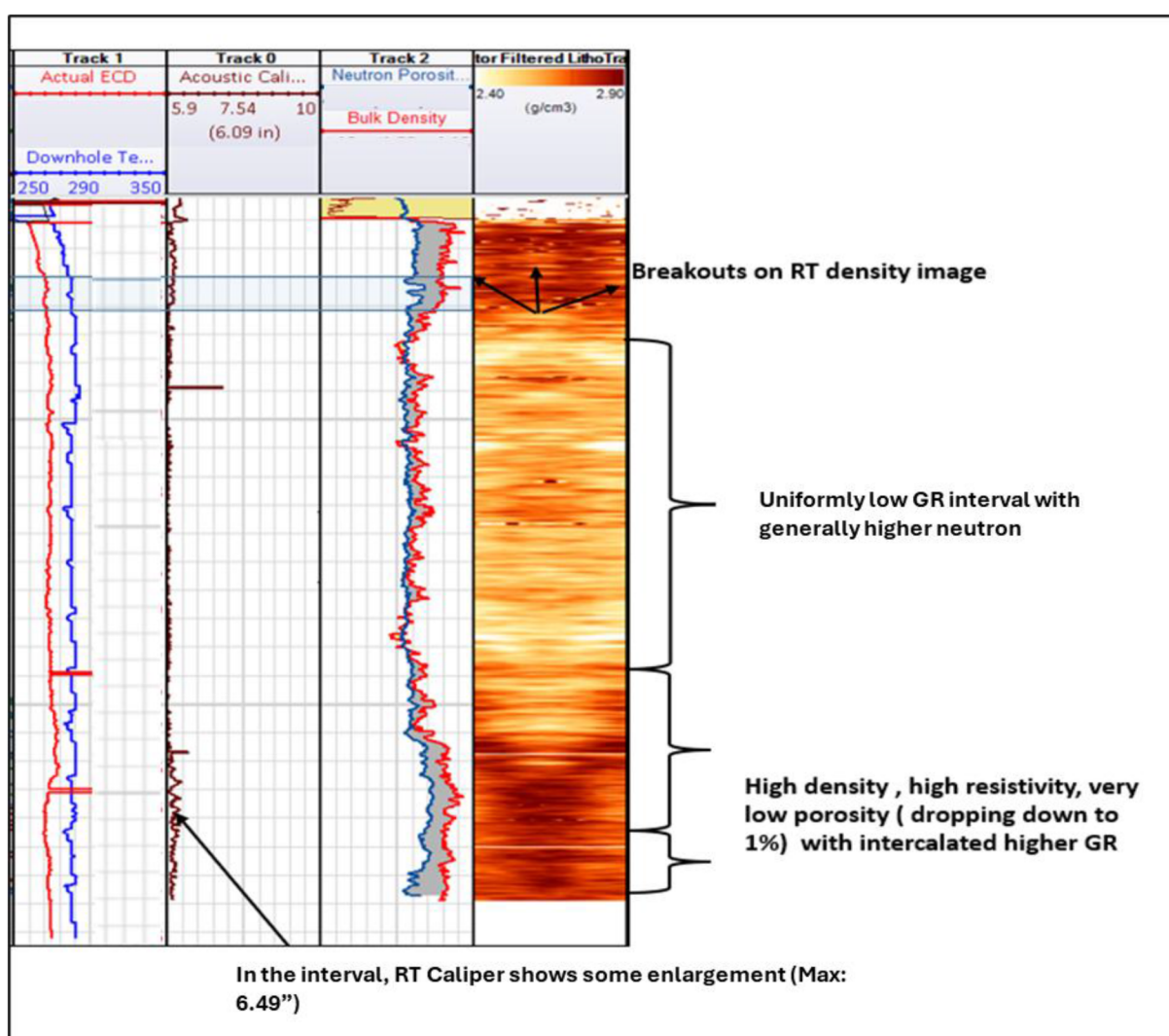


Figure 2—An example of composite plot monitor in Real Time highlighting any breakouts on density image, enlargement on acoustic caliper, change in lithology, and porosity to alert the operator; thus, helping directly for proactive decision making in favor of for example taking extra caution during pipe connection, optimized hole cleaning, optimizing mud parameters such as bridging material to mitigate any wellbore instability challenges.

Results & Conclusions

A long horizontal lateral section exceeding 5,000 feet was successfully drilled and completed while maintaining a consistent and stable mud weight throughout the operation. Notably, the entire drilling process was executed without any wellbore instability-related Non-Productive Time (NPT). The wellbore remained smooth and intact all the way to the planned Total Depth (TD), underscoring the effectiveness of the integrated planning and execution strategy.

This achievement was the result of a comprehensive and collaborative approach, combining a meticulously conducted pre-drill GeoMechanical Study and Wellbore Stability Analysis (WBS) with Real-Time Geomechanics Services. These efforts were further enhanced by the strategic deployment of Logging While Drilling (LWD) tools—particularly density imaging and acoustic caliper tools—which enabled proactive, real-time decision-making. This allowed the drilling team to swiftly respond to evolving subsurface conditions, ensuring continuous operational efficiency.

In alignment with regional tectonic characteristics, the well was drilled in the direction of the minimum horizontal stress (S_{hmin}) within a strike-slip faulting regime. In such settings, borehole breakouts typically manifest along the top and bottom of the wellbore. However, by carefully aligning the well trajectory with the regional stress orientation—confirmed during drilling—the team was able to mitigate these risks and maintain borehole integrity.

Drilling horizontal wells in strike-slip tectonic environments presents unique challenges, particularly when the trajectory aligns with S_{hmin} . These challenges include accurately predicting the optimal mud weight required to maintain wellbore stability, especially in the presence of reservoir heterogeneity and variable formation pressures. Such variability increases the risk of differential sticking, particularly in high-permeability zones where elevated mud overbalance can narrow the safe mud weight window.

To address these complexities, the project relied on a robust pre-drill geomechanical and wellbore stability model, which served as the foundation for determining the appropriate mud weight to initiate drilling safely. However, recognizing the inherent uncertainties in subsurface conditions, real-time geomechanics monitoring was employed as a critical risk mitigation tool. By continuously analyzing downhole data and dynamically updating the geomechanical model, the drilling team was empowered to make informed, proactive decisions that minimized risk and prevented NPT.

The success of this well exemplifies the value of integrating advanced geomechanical modeling with real-time monitoring and decision-making. It highlights how a data-driven, collaborative approach can significantly enhance drilling performance, reduce operational risks, and ensure the safe and efficient delivery of complex well trajectories.

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